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Impact of the menstrual cycle phases and time of day on markers of stress: salivary α-amylase and secretory immunoglobulin A

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ABSTRACT

Previous reports suggest that the menstrual cycle (MC) phases can impact cortisol concentrations. However, research is needed on whether the MC impacts other markers of stress and immune function. It has also been shown that some biomarkers are impacted by time of day, although differences between morning (AM) and afternoon (PM) biomarkers have not been studied over the course of the MC. This study assessed the effect of MC phases and time of day on salivary stress biomarkers [salivary α-amylase (sAA), secretory immunoglobulin A (SIgA)], progesterone, resting blood pressure and resting heart rate (RHR). A single-group repeated measure design was employed in which seventeen participants (n = 17) monitored their MC for two months while attending eight experimental sessions which included both AM and PM sessions during each predicted 1) menses, 2) follicular, 3) ovulatory and 4) luteal phases. Resting blood pressures, heart rates, body composition parameters (assessed via bioelectrical impedance analysis), sAA and SIgA concentrations were assessed. No time of day x MC phase interactions ($p > 0.05$) were noted for sAA or SIgA, resting blood pressure, heart rate, or body composition parameters. However, sAA and RHR were significantly higher in the PM, while SIgA was significantly higher in the AM. These data suggest that the MC phases do not impact sAA or SIgA, resting blood pressure, heart rates, or body composition parameters. However, time-of-day impacts RHR and concentrations of sAA and SIgA. These findings provide implications for female participants in research dealing with these biomarkers.

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Females; menstruation; hypothalamic-pituitary-adrenal axis; sympathoadrenal axis; menses; women

1. Introduction

Historically, females have been excluded from clinical trials and exercise science-related research due to the potential impact of the menstrual cycle (MC) on physiological biomarkers and performance parameters (Costello et al., 2014; Cowley et al., 2021; Liu & Mager, 2016; McNulty et al., 2024; Meignié et al., 2022). The underrepresentation of females in research trials is especially prominent in exercise/sports science and performance-related research and is typically attributed to fluctuations in sex hormones, such as estrogen and progesterone, which can impact stress responsiveness. Females have been shown to demonstrate reduced activation from the hypothalamic-pituitary-adrenal (HPA) and sympatho-adrenomedullary (SAM) axes (Reschke-Hernández et al., 2017; Teo et al., 2023; Verma et al., 2011). While still unclear, this is likely related to differences in the sensitivity of the glucocorticoid and mineralocorticoid receptors in the HPA axis (Teo et al., 2023). However, little attention has been given to the impact of the MC on other biomarkers of stress and immune system activation. This is an essential consideration, as elucidation on whether the MC impacts markers reflecting the SAM axis and immune system is necessary to establish a frame of reference concerning stress biomarker monitoring across the MC.

Biomarker monitoring has garnered attention among physiologists and recreationally active populations to assess responses to stressors and gauge overall performance or cardiometabolic health status (Sim et al., 2022). However, limited data exist regarding changes in biomarkers across the MC (Boisseau et al., 2013; Duchesne & Pruessner, 2013; Kirschbaum et al., 1999; Montero-López et al., 2018; Park & Watanuki, 2005; Quinn et al., 2021). Most studies have reported changes in cortisol across the MC phases (Hamidovic et al., 2020; Kulzhanova et al., 2023; Maki et al., 2015; Montero-López et al., 2018). Cortisol is one of the primary biomarkers of interest in assessing an individual's stress level as it reflects activation of the HPA axis (Ahmed et al., 2023; Huang et al.,

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2013). Stress exposure also activates the SAM axis (Goldstein, 2010) and the immune system (Mifsud & Reul, 2018; 2018) resulting in increased concentrations of blood catecholamines, salivary α-amylase (sAA), and secretory immunoglobulin A (SIgA) (Chojnowska et al., 2021; Cozma et al., 2017; Noushad et al., 2021). Catecholamines, sAA, SIgA, and sex hormones such as progesterone also have well-documented diurnal cycles (Cajochen et al., 2022; Caufriez et al., 2018; Fujimoto et al., 1990; Hucklebridge et al., 1998; Rohleder et al., 2004; Shirakawa et al., 2004). In young, healthy males and females sAA demonstrates an inverse circadian rhythm to SIgA and Cortisol (Nater et al., 2007; Rohleder et al., 2004). SIgA and cortisol follow a circadian rhythm that peaks in the morning and decreases throughout the day (Cajochen et al., 2022; Hucklebridge et al., 1998; Shirakawa et al., 2004). However, there is a need to further understand the impact of MC phases (Hucklebridge et al., 1998; Shirakawa et al., 2004). Cajochen et al. (2022), controlled for the MC phase by having all their young (19-35 year old) female participants complete the study during their luteal phase, however research examining the impact of the phases of the MC (and time of day) is needed.

In addition, it has been suggested that female's experience fluctuations in body mass throughout their MC that has been attributed to shifts in fluid retention (Tomazo-Ravnik & Jakopič, 2006; White et al., 2011). This is commonly associated with objective measures, such as feelings of being bloated during the onset of menstruation (Hicks et al., 2017; White et al., 2011). However, the relationship between fluid fluctuations through the MC and other markers of body composition are not fully understand. For instance, some work has shown differences in extracellular body water, total body water and body mass between the follicular and luteal phase (Kanellakis et al., 2023), while other findings have only shown changes in extracellular body water with no impact on body mass between the MC phases (Hicks et al., 2017). It should be noted that earlier studies assessing the relationship between body composition and various MC phases were utilizing a single frequency (e.g. 50-100 kHz) bioelectrical impedance analyzer (BIA) (Daniusevičiūtė et al., 2018; Gualdi-Russo & Toselli, 2002; Tomazo-Ravnik & Jakopič, 2006) while recent advancements in BIA technology utilize multiple frequencies that reach up to 1000 kHz that is likely more sensitive to fluctuations (Cumberledge et al., 2018; Hicks et al., 2017; Kanellakis et al., 2023). With this in mind, to further elucidate the impact of four different phases of the MC on various composition markers (i.e., body mass, total body water, extracellular and intracellular water, fat mass, fat free mass, and body fat percent) we included bioelectrical impedance analysis (BIA) as a secondary measure in the current study.

There is a need to expand our understanding of female physiology and the impact of the MC on various health and performance parameters. Therefore, the primary purpose of the present study was to determine if the MC phases (i.e., menses, follicular, ovulation, and luteal) influence salivary stress biomarkers sAA and SIgA. A secondary purpose of the study was to assess if body composition or resting blood pressure and heart rate were influenced across the various MC phases.

2. Materials and methods

2.1. Participants and experimental design

This study utilized a single-group repeated measures design to examine potential changes in stress markers and body composition across the MC phases. In addition, the same outcomes were compared between morning and evening time points. Fifty-four ($n = 54$) healthy, college-aged female participants were recruited primarily via flyers, verbal recruitment, and emails from the University campus to participate. Participants were required to 1) be between 18-40 years old, 2) eumenorrheic (MC between 21- 35 days in length) (Smith et al., 2022), 3) remain free from any signs, symptoms, and diagnosis of cardiovascular and metabolic disease(s) such as high blood pressure, heart disease, peripheral circulatory disorder, high blood cholesterol or triglyceride levels, thyroid, pancreas, liver or kidney dysfunction, diabetes or blood sugar abnormality, 4) remain free from any tobacco product use, and 5) refrain from any medication and dietary supplementation that may interfere with any of the outcome study variables (anxiety, depression medication/supplementation). In addition, participants were excluded from partaking in the study if any of the following occurred within three months of study initiation: an abortion or miscarriage, irregular period (one or more missed periods within the last 3 months), use of emergency contraception (e.g. Plan B), or rare adverse life event (e.g. divorce, loss of a loved one, emergency medical event). Participants who were pregnant or trying to become pregnant were not included. Participants were excluded during the study for having an irregular cycle if their experimental menstruation did not arrive within 5 days of the predicted date, or occurred outside of 21-35 days of their baseline menstruation. This model was based on suggestions by Elliott-Sale et al. (2021) for female participants in sport research. Lastly, participants who were taking hormonal contraceptives were excluded from the study. Prior to the initiation of the study, each participant was familiarized with the study protocol and procedures and provided verbal and written informed consent to participate in the study. The participants completed a health history questionnaire during the initial visit. The testing sessions occurred in the morning (AM) between 0500 and 1000 and in the evening (PM) between 1400 and 1900 on the same day of each respective phase: menses, follicular, ovulatory, and luteal. Participants were asked to fast for ≥8 hours before the AM testing sessions and ≥1 hour before PM testing sessions. Participants were asked to maintain their current diet and routines throughout the study. Prior to all testing sessions, participants were asked to refrain from strenuous exercise for 3 hours, any alcohol consumption for 48 hours and any nonsteroidal anti-inflammatory drugs for 24 hours. All experimental procedures were approved by the Institutional Review Board of Texas State University (IRB#: 7892) and were conducted in accordance with the Declaration of Helsinki.

2.2. Participant demographics

A total of seventeen ($n = 17$) female participants (age = 21.6 ± 2.2 years; Height = 160.3 ± 6.2 cm; Body mass = 62.1 ± 15.6 kg; BMI = 24.7 ± 4.6 kg/m^2) completed the study and were included in the statistical analysis. Initially, fifty-four females were recruited; however, sixteen females did not meet the study inclusion criteria, and three females declined to participate. Upon beginning the experimental procedures, six females no longer fit inclusion criteria for the study, wherein four had changed or started new medications, two experienced a stressful life event (i.e., wisdom tooth surgery and concussion). One participant was unable to complete all phases of testing. Another five females were excluded during the study due to irregular MCs.

2.3. Testing overview

Each testing session occurred at the Metabolic and Applied Physiology Laboratory at Texas State University (San Marcos, TX, USA). Participants reported to the laboratory for AM and PM testing sessions on four separate occasions (i.e., two sessions/day for four days for a total of 8 sessions) in alignment with their MC phases. This included the following four phases: 1) menses, 2) follicular, 3) ovulatory, and 4) luteal. All AM and PM sessions for a respective phase occurred on the same day. The participants utilized the Ovia: Fertility, Cycle, Health smartphone app (Ovia Health, Boston, MA, USA) to determine and monitor the four MC phases for one month prior to testing to establish a baseline and to determine appropriate experimental testing days. Tracking the MC for 1 month or more prior to testing has been established as acceptable methodological design (Smith et al., 2022). Upon arrival to the testing sessions, participants had body composition assessed and subsequently completed a mouth rinse with bottled water. Participants then observed a 7-minute rest period prior to blood pressure and heart rate collection. Saliva sample collection occurred immediately following the blood pressure and heart rate assessments. During the AM sessions only, anthropometric and body composition measures were assessed. Resting blood pressures, heart rates measures and salivary biosamples were collected during the AM and PM sessions. The first AM (Visit 1) and PM testing sessions (Visit 2) were standardized to begin during the menses phase of the MC. The second two testing sessions (Visit 3 and Visit 4) occurred during the middle of the follicular phase (i.e., within a 4- to 7-day range during this phase based on the participant). The third two testing sessions (Visit 5 and Visit 6) occurred on the third or fourth day of the ovulatory phase. Lastly, the fourth two testing sessions (Visit 7 and Visit 8) occurred on the sixth day of the luteal phase. If the participants could not meet on the scheduled testing session date/time or within a 2-day window (i.e., before or after), they were disqualified from the study. Figure 1 displays the study timeline and testing sequence for the morning and evening testing sessions.

2.4. Experimental procedures

2.4.1. Menstrual cycle monitoring

The Ovia: Fertility, Cycle, Health smartphone app (Ovia Health, Boston, MA, USA) was used to determine the timing of the participant's MC phases. The participants monitored their MC for one month prior to attending the eight testing sessions. This consisted of participants entering daily whether they were currently menstruating into the Ovia app. Based on the baseline menses and onset of the experimental menses, Ovia provided an estimated timeline of menses and a predicted

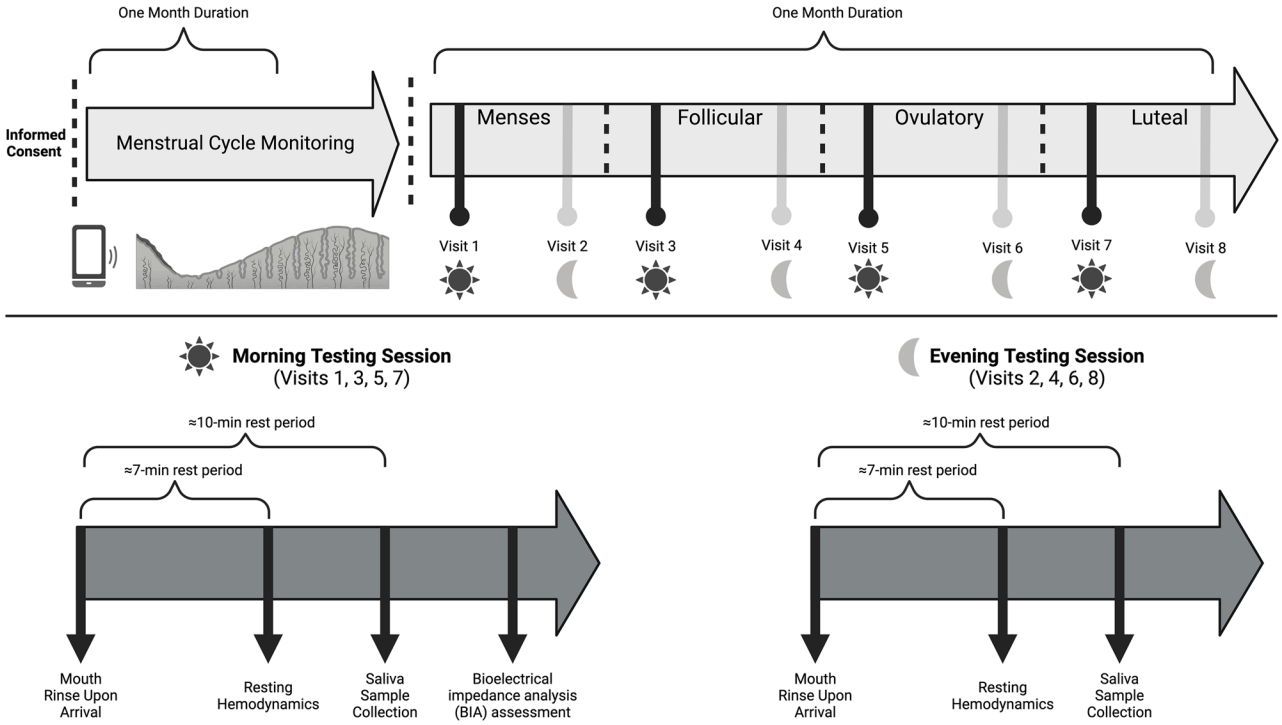

Figure 1. Study timeline and morning and evening testing session sequence. Note figure was created using biorender.com.

ovulatory phase based on back counting from the next predicted menstrual onset (a 6-day window 11 days before the onset of the next predicted menses). The time between menses and the predicted ovulatory phase was classified as the follicular phase, while the time between the predicted ovulatory phase and the next predicted menses was classified as the luteal phase. Once the baseline dates for the MC phases were established, the participants completed a second month of monitoring their MC with the Ovia app while attending the eight experimental testing sessions.

2.4.2. Resting blood pressures and heart rates

Resting blood pressures and heart rates were assessed in a seated position following a 7-minute rest. A portable Omron wrist blood pressure monitor (Omron; Kyoto, Japan) was used to determine resting heart rate (RHR), systolic blood pressure (SBP), and diastolic blood pressure (DBP) (Menezes et al., 2010; Moon et al., 2020; Saito et al., 2019). Two measures were recorded and averaged. However, if the first two measures were greater than 5 mmHg apart, a third measure was taken and included in the average.

2.4.3. Anthropometrics and body composition

Height was measured with a portable stadiometer (SECA, CA, USA). Body composition was assessed via an InBody BIA (Seoul, South Korea) which provided the following body composition parameters: total body water (TBW), fat mass (FM), fat-free mass (FFM), body fat percentage (BF%), and body mass (BM). In addition, the InBody calculated the participant's body mass index (BMI). Prior to each InBody assessment, participants were instructed to wear minimal clothing and were given an InBody wipe to cleanse their hands and bare feet. During the test, participants followed the audio cues of the InBody while being prevented from viewing the monitor screen. Participants remained blinded from their body composition data for the entire study duration.

2.4.4. Saliva sample collection and analysis

A passive drool collection method was utilized to obtain saliva samples from the participants. The participants were asked to perform a mouth rinse with bottled water ≈10 minutes prior to providing their saliva sample. After 10 minutes, the participants were asked to tilt their head forward, allow saliva to "pool" in their mouth, and passively transfer the saliva into a 2.0 mL micro-collection tube using a collection aid (Salimetrics, PA, USA). A total of ≈1.5 mL of saliva was collected per visit. Following collection, saliva samples were immediately stored at −80 °C until subsequent analysis.

Prior to analysis, saliva samples were thawed and centrifuged at 1500 x g for 15 minutes at 4 °C. The saliva samples were analyzed in duplicate for concentrations of sAA, SIgA, estradiol, and progesterone using commercially available enzyme-linked immunosorbent assay (ELISA) kits (Salimetrics, PA, USA). Product (kit) numbers for the ELISAs are as follows: progesterone: #1-1502; estradiol: #1-3702; sAA: #1-1902; SIgA: #1-1602. The saliva samples were shipped overnight, on dry ice, to a commercial laboratory to analyze sAA and SIgA (Salimetrics, PA, USA). The saliva samples were then returned to the Metabolic and Applied Physiology Laboratory for subsequent analysis of salivary estradiol and progesterone. A colorimetric plate reader (BioTek, Winooski, VT, USA) was used to determine absorbance. An automated washer was used for assays that required washing (BioTek, Winooski, VT, USA). The intra-assay and inter-assay coefficient of variation for these assays was <9%.

2.5. Statistical analysis

Statistical analysis was conducted via SAS v 9.4 (Cary, NC, USA). Two-way (time × phase) repeated measures analysis of variance (ANOVA) was used to examine potential time × phase interactions, as well as main effects for time (between AM and PM sessions) and phase (across the MC phases) for the salivary biomarkers (sAA, SIgA, estradiol, progesterone) and resting blood pressure and heart rate. One-way repeated measures ANOVAs were used to assess changes in body composition parameters across all MC phases. If an interaction or main effect was found ($p < 0.05$), Fisher's Least Significant Difference (LSD) *post hoc* tests were conducted to compare means.

3. Results

3.1. Menstrual cycle

The average cycle length was 28.5 ± 2.1 days, whereas each of the four individual phases began on the following day within the cycle: the menses phase began on day 1, the follicular phase began on day 6.1 ± 1.1, the ovulatory phase began on day 12.8 ± 2.6, and the luteal phases began on day 19 ± 2.9. Testing of each phase occurred on the following cycle days: the menses phase was tested on day 3.0 ± 1.2, the follicular phase was tested on day 9.0 ± 1.6, the ovulatory phase was tested on day 15.2 ± 2.1, and the luteal phase was tested on day 23.5 ± 2.5.

3.2. Resting blood pressure and heart rate

Table 1 displays resting blood pressure values across the MC phases. No time × phase interaction or main effects were found for SBP or DBP ($p > 0.05$; Table 1). In addition, no significant time × phase interaction or main effect for phase was found for RHR ($p > 0.59$; Figure 2). A main effect for time was noted ($F = 5.86$, $p = 0.017$), wherein *post hoc* analysis revealed RHR values were significantly higher during the PM compared to AM testing sessions (Figure 2).

3.3. Anthropometrics and body composition parameters

Table 2 presents all body composition parameters, including TBW, FFM, FM, BF%, and BM. No differences were found ($p > 0.05$) across MC phases for any body composition variable.

Table 1. Resting blood pressure results.

Variable	Session	Menstrual cycle phase				*p*-value		
		Menses	Follicular	Ovulatory	Luteal	T x P	T	P
Systolic BP (mmHg)	AM	113 ± 9	109 ± 7	109 ± 9	111 ± 7	0.26	0.20	0.48
	PM	112 ± 9	112 ± 8	112 ± 8	111 ± 7			
Diastolic BP (mmHg)	AM	75 ± 7	73 ± 5	73 ± 5	72 ± 6	0.80	0.35	0.50
	PM	75 ± 8	74 ± 8	74 ± 5	74 ± 5			

Data are expressed as means ± standard deviations. AM: Morning testing sessions; PM: Evening testing sessions; T x P: *p*-value for time x phase interaction test; T: *p*-value for time main effect; P: *p*-value for phase main effect; BP: blood pressure. Comparisons made via two-way repeated measures ANOVA with Fisher's LSD *post hoc* analysis. No time x phase interaction effects, or main effects for phase or time were noted.

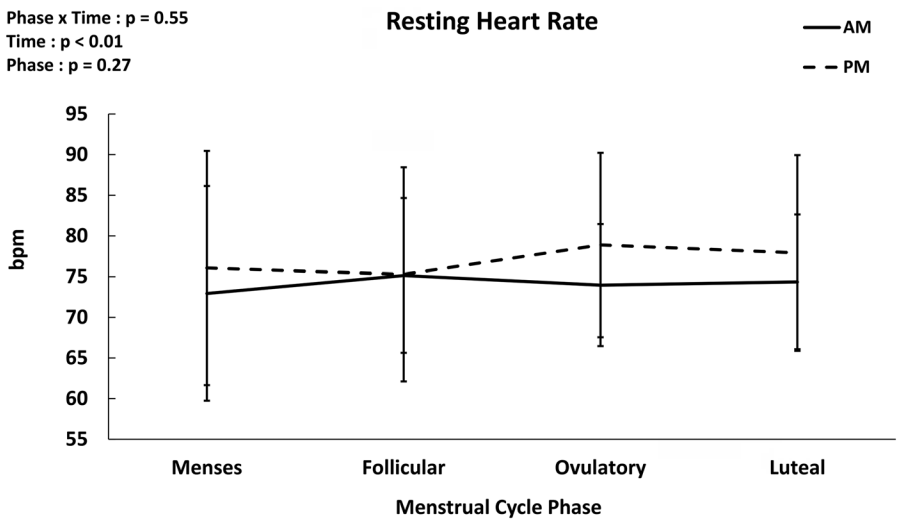

Figure 2. Values represent mean ± standard deviation. AM: morning testing session; PM: evening testing session. Note, significantly higher resting heart rate values (p = 0.017) were found during the evening testing sessions compared to the morning testing sessions.

Table 2. InBody body composition results.

Variable		Menstrual cycle phase				
		Menses	Follicular	Ovulatory	Luteal	*p*-value
Body Weight	kg	62.1 ± 15.6	63.4 ± 13.2	63.4 ± 12.7	63.3 ± 13.2	0.70
Body Fat	%	31.9 ± 8.3	31.4 ± 8.1	31.4 ± 8.1	31.4 ± 8.1	0.33
Fat Mass	kg	21.0 ± 9.0	20.5 ± 8.5	20.4 ± 8.5	20.5 ± 8.5	0.09
Fat Free Mass	kg	42.8 ± 6.8	42.9 ± 7.4	42.9 ± 7.1	42.8 ± 7.4	0.98
Water Weight	kg	31.3 ± 5.0	31.4 ± 5.4	31.4 ± 5.1	31.3 ± 5.4	0.98

Data are expressed as means ± standard deviations. Comparisons made via one-way ANOVA.

3.4. Salivary stress biomarkers

Figure 3 presents mean sAA and SIgA concentrations across the MC phases during the AM and PM testing sessions. No time × phase interaction effects were found for sAA ($p = 0.54$) or SIgA ($p = 0.94$). Main effects for time were found for sAA ($F = 154.54$, $p < 0.0001$) and SIgA ($F = 4.68$, $p = 0.032$). *Post hoc* analysis revealed that sAA concentrations were significantly higher during the PM testing sessions compared to AM (Figure 3a). Conversely, concentrations of SIgA were significantly higher in the AM compared to PM testing sessions (Figure 3b).

3.5. Salivary hormones

Figure 4 presents mean salivary progesterone and estradiol concentrations across the MC phases during the AM and PM sessions. No time × phase interaction effects were found for salivary progesterone ($p = 0.89$) or estradiol ($p = 0.53$). Main effects for time ($F = 11.65$, $p = 0.0009$) and phase ($F = 11.91$, $p < 0.0001$) were found for progesterone, while no main effects for time or phases were found for estradiol. *Post hoc* analysis revealed that salivary progesterone concentrations were significantly higher during the AM testing sessions compared to PM, and concentrations were higher during the luteal phases than all other phases (Figure 4a). Salivary estradiol concentrations were not different ($p > 0.05$) between the AM and PM sessions nor between phases (Figure 4B).

4. Discussion

This study aimed to determine the impact of the MC on salivary biomarkers of stress (i.e., sAA and SIgA), body composition parameters, and resting blood pressure and heart rate among eumenorrheic females. Our findings align with previous research that documented the diurnal profile of sAA, such that sAA concentrations are typically higher in the PM compared to AM (Nater et al., 2007). Resting heart rate and sAA concentrations in the present study were higher during the PM compared to the AM testing sessions, which mirrors

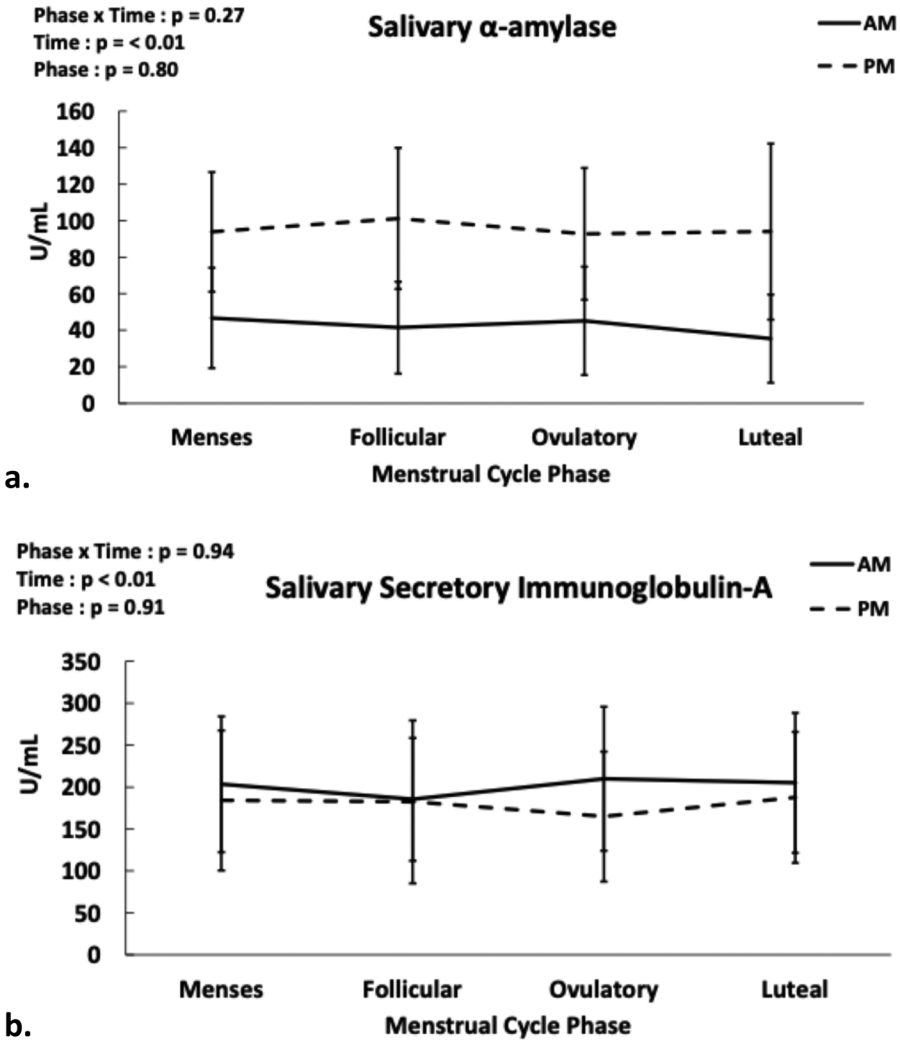

Figure 3. Values represent mean ± standard deviation. (a) Represents salivary α-amylase, and (b) represents salivary secretory immunoglobulin A. AM: morning testing session; PM: evening testing session. Note: salivary α-amylase concentrations were higher ($p < 0.0001$) during the PM testing sessions compared to AM (a). Salivary secretory immunoglobulin A concentrations were higher ($p = 0.032$) in the AM testing sessions compared to PM (b). Note the number of observations (n) may vary due to missing or inadequate samples: For salivary α-amylase and secretory immunoglobulin A: n = 14 - 17 for AM/PM phases.

the previously delineated diurnal sAA patterns noted among females (Nater et al., 2007; Nater & Rohleder, 2009; Rohleder et al., 2004). Concentrations of SIgA were higher during the AM compared to PM testing sessions, although the diurnal profiles for SIgA are less understood. This is the first study to investigate the potential impact of the MC phases on AM and PM salivary stress biomarkers. Taken together, these data suggest there are diurnal influences on concentrations of sAA and SIgA, as well as RHR values, among females; however, the phases of the MC do not impact sAA or SIgA, body composition, or resting blood pressure and heart rate parameters.

Females have long been excluded from clinical and scientific research due to the potential uncertainty of confounding effects from hormonal fluctuation across the MC (Kajantie & Phillips, 2006; Komesaroff et al., 1999; Liu & Mager, 2016; Reschke-Hernández et al., 2017; Teo et al., 2023; Uhl et al., 2007; Verma et al., 2011). Interestingly, males and females have been shown to respond differently to stressors (Reschke-Hernández et al., 2017; Teo et al., 2023; Verma et al., 2011). This may be related to hormonal differences (e.g. estradiol may blunt SAM response) (Kajantie & Phillips, 2006), reduced catecholamine levels and blood pressure with estrogen supplementation (Komesaroff et al., 1999), and/or different sensitivities of glucocorticoid/mineralocorticoid receptors (Teo et al., 2023). Furthermore, researchers have postulated that hormone fluctuations will impact female stress responsiveness (Verma et al., 2011). To date, few studies have explored the impact of the MC phases on SAM reactivity to stress. For example, Espin et al. (2019) compared sAA responses between males and females in response to a mental stressor and demonstrated that females in the follicular phase had a less pronounced stress-induced sAA response compared to males and females in the luteal phase. These findings from Espin et al. (2019) are interesting, since sAA reactivity may be impacted by estrogen or progesterone concentrations (Gordon & Girdler, 2014; Kajantie & Phillips, 2006). Moreover, progesterone levels differ between the luteal and follicular phases, wherein the highest levels are present during the luteal phase (Gordon & Girdler, 2014; Kajantie & Phillips, 2006), and it is postulated that progesterone may enhance SAM axis reactivity (thus resulting in higher sAA concentrations) (Espin et al., 2019). While the effects of

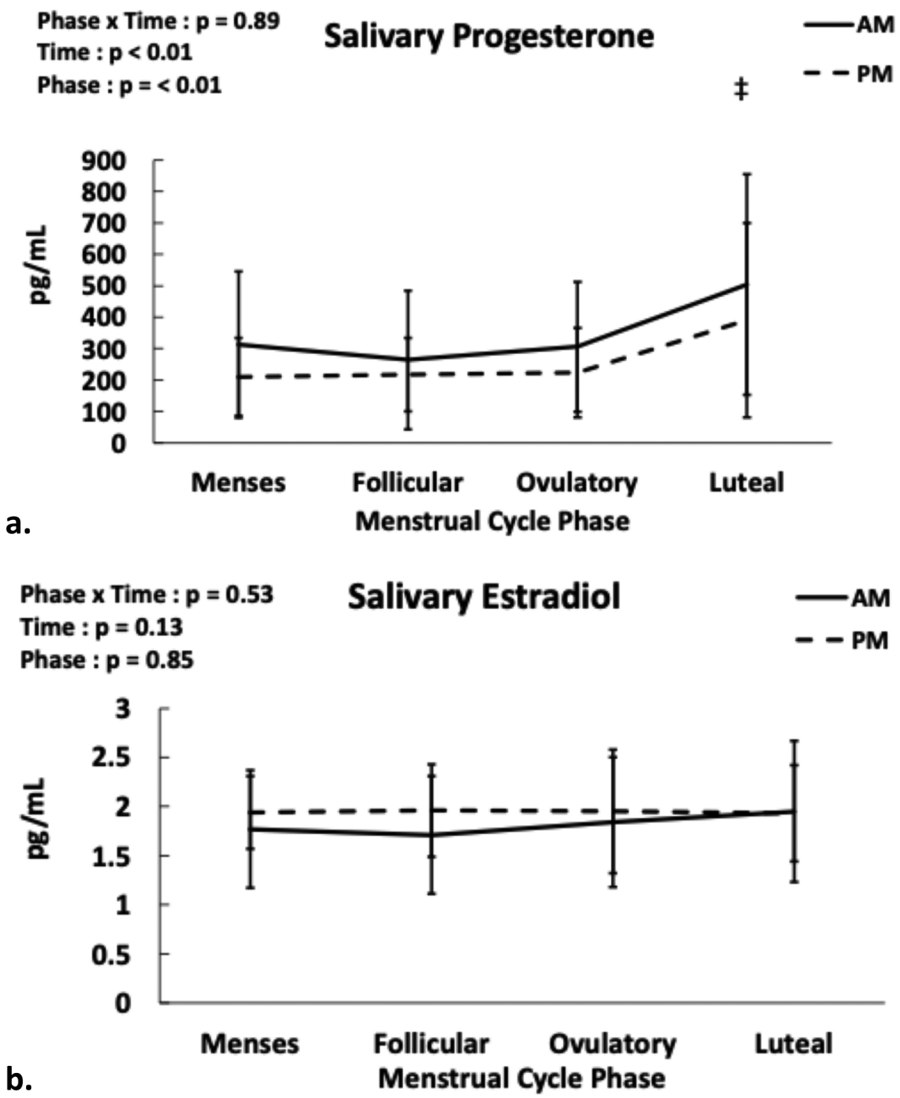

Figure 4. Values represent mean ± standard deviation. (a) Represents salivary progesterone, and (b) represents salivary estradiol. AM: morning testing session; PM: evening testing session. Note: salivary progesterone concentrations were higher ($p = 0.0011$) during the AM testing sessions compared to PM, and progesterone concentrations were higher during the luteal phases compared to all other phases (‡; $p < 0.001$) (a). Salivary estradiol concentrations were not different ($p > 0.05$) between the AM and PM sessions nor between phases (b). Note the number of observations (n) may vary due to missing or inadequate samples: For salivary progesterone and estradiol: n = 15 - 17 for AM/PM phases.

progesterone on the stress response remain unclear, our findings demonstrate that sAA concentrations are not influenced by the four primary phases of the MC. Finally, the present findings demonstrating diurnal changes in sAA concentrations (i.e., higher concentrations in the PM compared to AM) are consistent with previous studies (Nater et al., 2007; Rohleder et al., 2004) and are likely related to sAA's connection to the autonomic nervous system (Ali et al., 2020).

The findings regarding SIgA's diurnal profile (i.e., higher AM and lower PM concentrations) agree with previous reports (Cajochen et al., 2022; Hucklebridge et al., 1998; Shirakawa et al., 2004). However, the four MC phases did not impact salivary SIgA concentrations. Salivary SIgA has been utilized in various settings to assess the stress response and reflect the activation of the immune system. Importantly, SIgA is the host's first line of defense in combating foreign pathogens or invading organisms (Pietrzak et al., 2020). Previous work has reported that amenorrhoeic females express lower concentrations of salivary SIgA than their eumenorrheic counterparts, possibly due to a hormonal influence on immunoglobulin production and secretion (Gómez et al., 1993). Yet, a clear association between sex hormones and salivary SIgA concentrations has not been found (Zwahlen & Stute, 2024). Moreover, as noted in our study, previous data similarly suggest salivary SIgA concentrations are unchanged across the MC (Burrows et al., 2002; Gillum et al., 2014). However, it is important to note differences in methodological procedures since past research has assessed changes in SIgA concentrations in response to acute exercise during luteal and follicular phases (Gillum et al., 2014) or studied changes among highly trained athletes (Burrows et al., 2002).

In terms of implications for selections of salivary stress markers in future work dealing with female participants, sAA and SIgA are both suitable noninvasive biomarkers to assess acute stress. Salivary sAA and SIgA tend to be more responsive to acute stress as compared to salivary cortisol (McAllister et al., 2022, 2024; McAllister & Martaindale, 2021). Due to the slow acting nature of cortisol responses, it is likely that long

resting periods (~60 min) are needed to establish an appropriate baseline for cortisol concentrations prior to stress exposure. Salivary cortisol release involves a multi-step process starting with the release of corticotropin-releasing hormone and adrenocorticotropic hormone before stimulating the adrenal cortex to release cortisol into the saliva (Hellhammer et al., 2009). Increased concentrations of sAA and SIgA are measurable within minutes after exposure to a stressor due to being impacted by the sympathetic nervous system (SNS)(Nater & Rohleder, 2009). For example, past work by McAllister et al. (2022), demonstrated significant increases in sAA in response to an acute stressor (~2-3 minutes) with no measurable changes in salivary cortisol. However, cortisol is also likely more impacted by MC phase as compared to SIgA and sAA (Caufriez et al., 2018). Thus, the present findings suggest that markers of immune function and SAM axis activity are not impacted by MC phase when females are not exposed to a stressor.

Our findings align with past work, which demonstrated resting blood pressure and heart rate follow a circadian pattern, wherein heart rate and blood pressure increase toward the end of the day and fluctuate throughout the day depending on stress exposure (e.g. exercise, mental stress, etc.) (Veerman et al., 1995). To date, only a few studies have investigated the MC's influence on RHR and reported conflicting data. For example, reports noted higher RHR values during the luteal compared to the follicular phase (McKinley et al., 2009; Moran et al., 2000) while other reports found no difference in RHR across the MC phases (Kwissa et al., 2022; Tousignant-Laflamme & Marchand, 2009; Yildirir et al., 2002). This discrepancy is most likely from inconsistent methodological approaches. For instance, studies that found significant differences between menstrual phases used continuous (McKinley et al., 2009) or daily monitoring (Moran et al., 2000), or performed testing during two (luteal vs follicular) or three (luteal, ovulatory, follicular) phases per MC (Kwissa et al., 2022; Tousignant-Laflamme & Marchand, 2009; Yildirir et al., 2002). The current study measured RHR, SBP, and DBP twice per phase (AM and PM) and included four phases (menses, follicular, ovulatory, luteal) of the MC. Regarding blood pressure, several studies have reported no differences across the MC phases, and our findings mirrored these reports (Arifuddin et al., 2012; Eagan et al., 2022; Gordon et al., 2018; Kwissa et al., 2022).

Regarding body composition parameters, typically it is recommended when assessing body composition in females, measures should not be collected during their menstruation phase due to fluid retention (Wagner & Heyward, 1999). However, recent findings have suggested that the MC does not impact body composition measures (Cumberledge et al., 2018; Gould et al., 2021; Kanellakis et al., 2023; Kuikman et al., 2024). Gould et al. (2021), demonstrated that body composition parameters, assessed via a four-compartment model and DEXA, were not affected by the MC's early follicular nor mid-follicular phases. These findings are similar to more recent findings by Kuikman et al. (2024) who demonstrated no significant differences in body composition between different phases of the MC or between hormonal and non-hormonal contraceptive users. Our research team assessed the four phases of the MC and corroborated the findings of Gould et al. (2021) and others (Cumberledge et al., 2018; Hicks et al., 2017) utilizing the BIA. These findings suggest that phases of the MC do not impact body composition measures determined by the BIA.

Regarding the salivary sex hormones, salivary progesterone has been previously shown to correlate with plasma concentrations (Vining & McGinley, 1987). The present findings demonstrated that progesterone concentrations were higher during the luteal phase compared to all other phases. The observed changes in progesterone concentrations align with previous work of the MC cycle and indicate the shift into the luteal phase (Reed & Carr, 2000; Thiyagarajan et al., 2024). Concentrations were also higher in the AM compared to PM. The diurnal variation of progesterone found in the current study has been noted in previous research (Caufriez et al., 2018) and is impacted by menstrual status (Fujimoto et al., 1990; van Kerkhof et al., 2015). Salivary concentrations of estradiol were not different between AM and PM sessions or MC phases, aligning with previous research (van Kerkhof et al., 2015). While past studies have demonstrated changes in salivary concentrations of progesterone and estradiol across the MC (Chatterton et al., 2005; Gandara et al., 2007; Wong et al., 1990), recent findings have suggested that salivary estradiol may be a poor predictor of the MC phase (Arslan et al., 2023). As such, additional research is warranted to demonstrate a clear relationship between the MC phases and salivary sex hormones.

The present study has limitations that should be acknowledged. First, the participants were asked to track their MCs via the Ovia Health smartphone app for only one month before attending the experimental testing sessions; while this is acceptable, ≥3 months may be ideal for MC monitoring (Smith et al., 2022). Second, we were unable to confirm ovulation as no concurrent blood or urinary analysis was done, which is generally required to confirm the two distinct follicular and luteal phases. Third, all participants began testing at the onset of menses. Therefore, it is possible this was a factor that could impact stress measures. The study is also limited by a small sample size which should be viewed as a limitation. Lastly, no additional screening of daily stressors was included in this study. It is also important to note that we did not track dietary or physical activity patterns throughout the study, and these factors may have impacted the results. However, a strength of the study is that we included resting blood pressure, heart rate and salivary measures twice per day (both AM and PM) across all MCs.

5. Conclusion

The results of the present study suggest that stress biomarkers, sAA and SIgA, and salivary sex hormones estradiol and progesterone are not affected by the four MC phases (i.e., menses, follicular, ovulatory, and luteal). In addition, body composition parameters and resting blood pressure and heart rate unaltered by the MC phases. Rather, our findings demonstrate that sAA and SIgA follow distinct diurnal patterns among eumenorrheic females. Together, these data provide

implications for female involvement in clinical and scientific research dealing with these biomarkers. Future work should focus on the impact of the MC phases on stress biomarkers in response to acute, stressful conditions of varying intensity.

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Authors' contributions

Funding acquisition: MJM; conceptualization and research design: MJM and HSW; performed data collection: JKW; analyzed data: MJM; interpreted experiment results: JKW, CCD, DEG, HSW, and MJM; prepared figures: JKW, DEG, and MJM; drafted manuscript: JKW and CCD; edited and revised manuscript: DEG, HSW and MJM; and all authors approved the final version of the manuscript.

Institution and ethics approval and informed consent

All experimental produces subsequently described were approved by the Institutional Review Board of Texas State University (IRB: 7892) in line with the Declaration of Helsinki.

Disclaimers

The content is solely the authors' responsibility and does not necessarily represent the official views of Texas State University.

Disclosure statement

The authors have no conflicts of interest to declare.

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Data availability statement

Data will be made available upon reasonable request by contacting corresponding author. Note, consent was not obtained to make data publicly available.

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